Laporan Ujian Tengah Semester

Client Server



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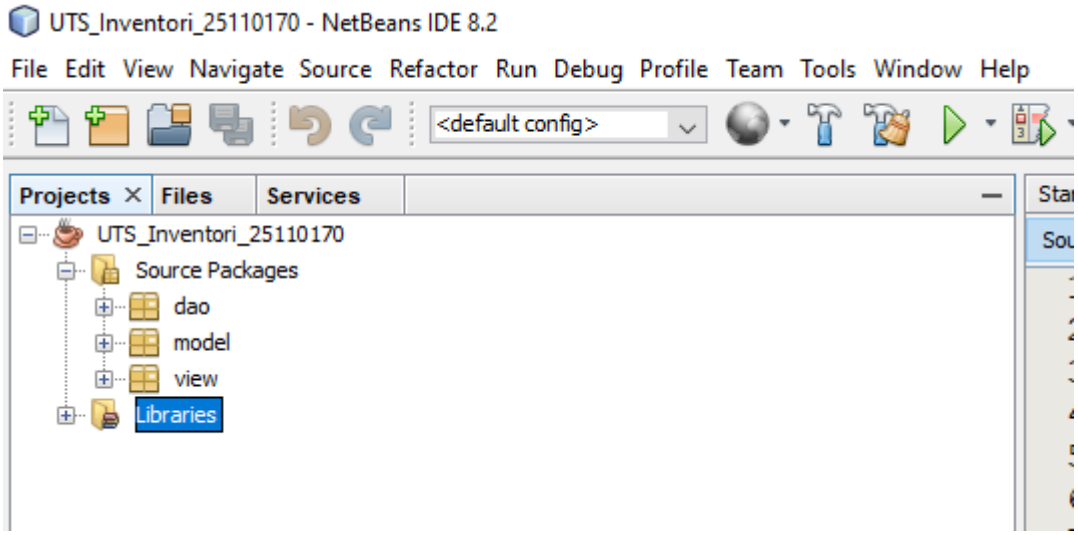
Mika lestari Nainggoloan

Nim:25110170

Universitas Minar Indonesia

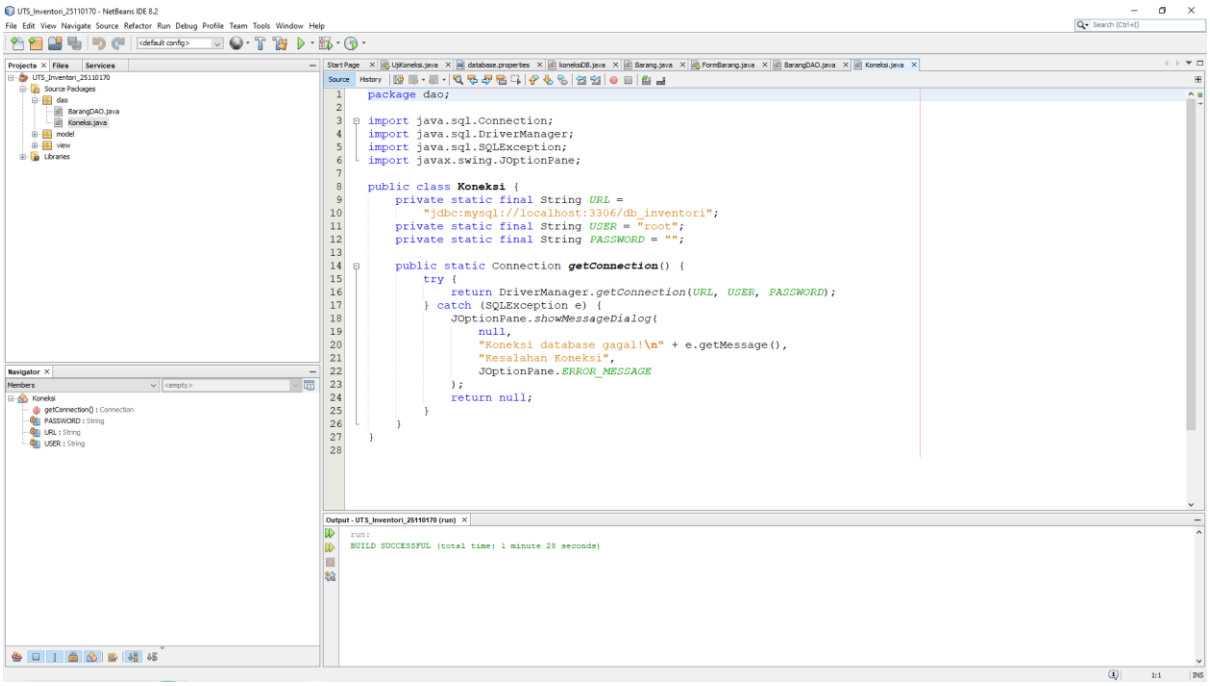
2026/2027

- Proyek



Berisi proyek netbeans dengan nama UTS_Inventori_2511017 dan 3 paket yakni dao, model dan view

- Class Koneksi



```
package dao;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import javax.swing.JOptionPane;

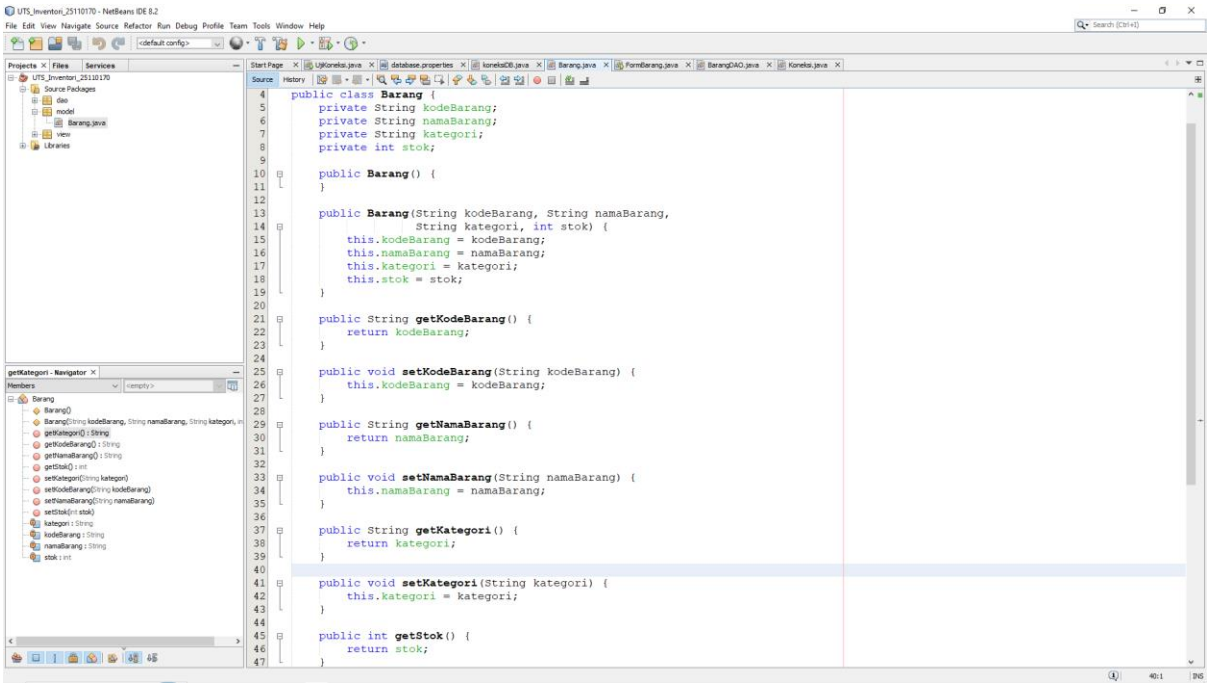
public class Koneksi {
    private static final String URL =
        "jdbc:mysql://localhost:3306/db_inventori";
    private static final String USER = "root";
    private static final String PASSWORD = "";

    public static Connection getConnection() {
        try {
            return DriverManager.getConnection(URL, USER, PASSWORD);
        } catch (SQLException e) {
            JOptionPane.showMessageDialog(
                null,
                "Koneksi database gagal!\n" + e.getMessage(),
                "Kesalahan Koneksi",
                JOptionPane.ERROR_MESSAGE
            );
            return null;
        }
    }
}
```

```
run:
BUILD SUCCESSFUL (total time: 1 minute 28 seconds)
```

Dalam class koneksi di paket dao yang berisi code untuk mengembalikan objek connection ke db_inventori.

- Class barang



```
public class Barang {
    private String kodeBarang;
    private String namaBarang;
    private String kategori;
    private int stok;

    public Barang() {
    }

    public Barang(String kodeBarang, String namaBarang,
                  String kategori, int stok) {
        this.kodeBarang = kodeBarang;
        this.namaBarang = namaBarang;
        this.kategori = kategori;
        this.stok = stok;
    }

    public String getKodeBarang() {
        return kodeBarang;
    }

    public void setKodeBarang(String kodeBarang) {
        this.kodeBarang = kodeBarang;
    }

    public String getNamaBarang() {
        return namaBarang;
    }

    public void setNamaBarang(String namaBarang) {
        this.namaBarang = namaBarang;
    }

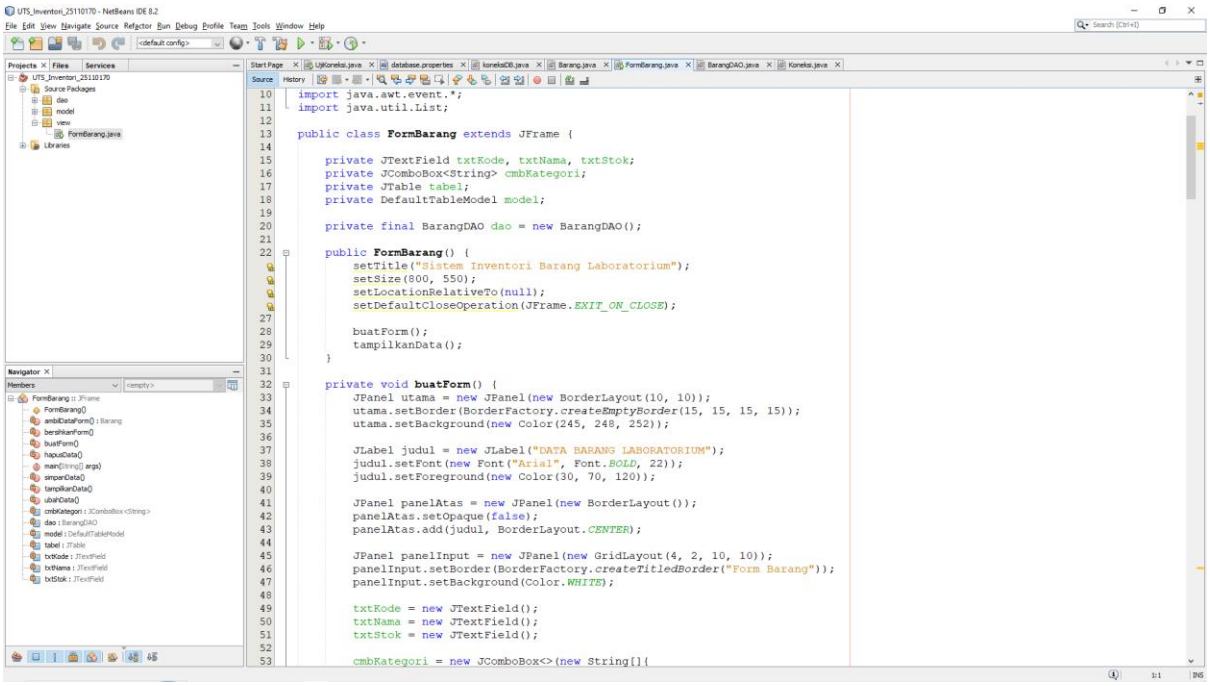
    public String getKategori() {
        return kategori;
    }

    public void setKategori(String kategori) {
        this.kategori = kategori;
    }

    public int getStok() {
        return stok;
    }
```

Class barang dalam paket model yang berisi atribut private, kodeBarang, namaBarang, kategori, stok, beserta constructor, getter dan setter.

- Form barang



```
import java.awt.event.*;
import java.util.List;

public class FormBarang extends JFrame {

    private JTextField txtKode, txtNama, txtStok;
    private JComboBox<String> cmbKategori;
    private JTable tabel;
    private DefaultTableModel model;

    private final BarangDAO dao = new BarangDAO();

    public FormBarang() {
        setTitle("Sistem Inventori Barang Laboratorium");
        setSize(800, 550);
        setLocationRelativeTo(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        buatForm();
        tampilkanData();
    }

    private void buatForm() {
        JPanel utama = new JPanel(new BorderLayout(10, 10));
        utama.setBorder(BorderFactory.createEmptyBorder(15, 15, 15, 15));
        utama.setBackground(new Color(245, 248, 252));

        JLabel judul = new JLabel("DATA BARANG LABORATORIUM");
        judul.setFont(new Font("Arial", Font.BOLD, 22));
        judul.setForeground(new Color(30, 70, 120));

        JPanel panelAtas = new JPanel(new BorderLayout());
        panelAtas.setOpaque(false);
        panelAtas.add(judul, BorderLayout.CENTER);

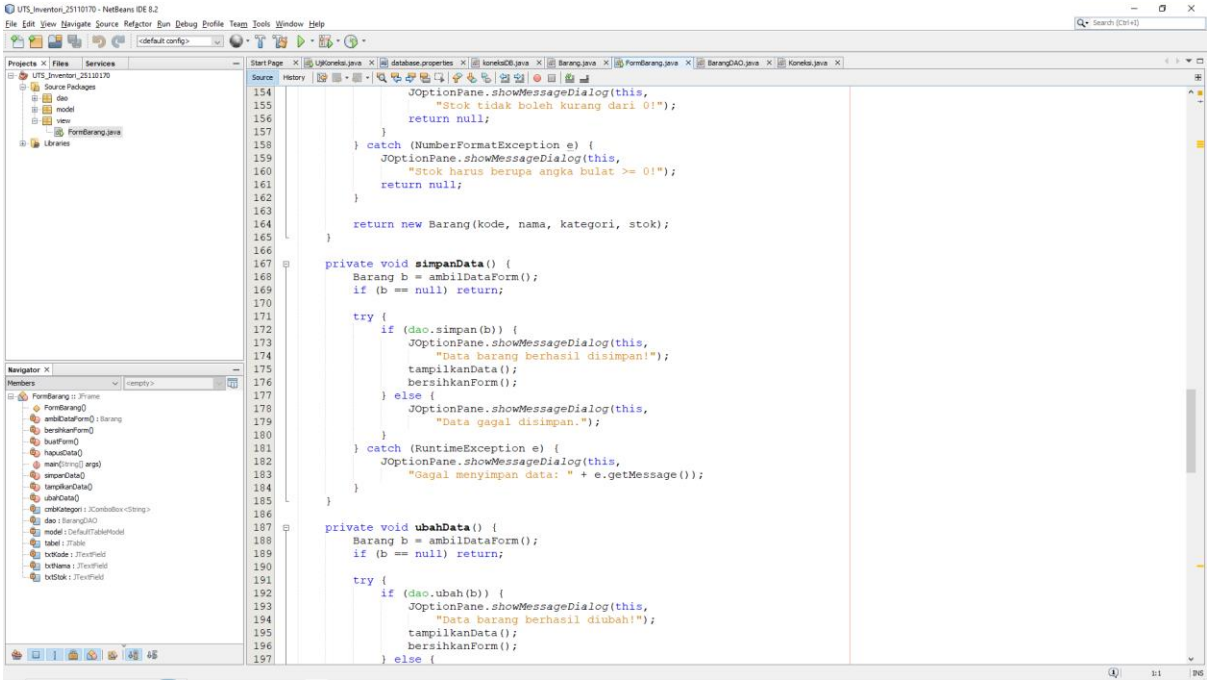
        JPanel panelInput = new JPanel(new GridLayout(4, 2, 10, 10));
        panelInput.setBorder(BorderFactory.createTitledBorder("Form Barang"));
        panelInput.setBackground(Color.WHITE);

        txtKode = new JTextField();
        txtNama = new JTextField();
        txtStok = new JTextField();

        cmbKategori = new JComboBox<>(new String[]{
```

Form barang di dalam paket view dengan JTextField untuk kode, nama dan stok. JComboBox untuk kategori. Beserta tombol simpan, ubah,hapus, dan bersihkan dan Jtable daftar barang.

- Tombol simpan



```
                    JOptionPane.showMessageDialog(this,
                        "Stok tidak boleh kurang dari 0!");
                    return null;
                }
            } catch (NumberFormatException e) {
                JOptionPane.showMessageDialog(this,
                    "Stok harus berupa angka bulat >= 0!");
                return null;
            }

            return new Barang(kode, nama, kategori, stok);
        }

        private void simpanData() {
            Barang b = ambilDataForm();
            if (b == null) return;

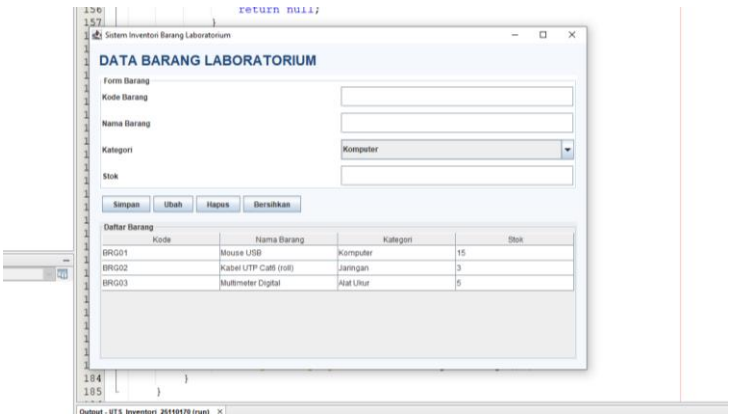
            try {
                if (dao.simpan(b)) {
                    JOptionPane.showMessageDialog(this,
                        "Data barang berhasil disimpan!");
                    tampilkanData();
                    bersihkanForm();
                } else {
                    JOptionPane.showMessageDialog(this,
                        "Data gagal disimpan.");
                }
            } catch (RuntimeException e) {
                JOptionPane.showMessageDialog(this,
                    "Gagal menyimpan data: " + e.getMessage());
            }
        }

        private void ubahData() {
            Barang b = ambilDataForm();
            if (b == null) return;

            try {
                if (dao.ubah(b)) {
                    JOptionPane.showMessageDialog(this,
                        "Data barang berhasil diubah!");
                    tampilkanData();
                    bersihkanForm();
                } else {
```

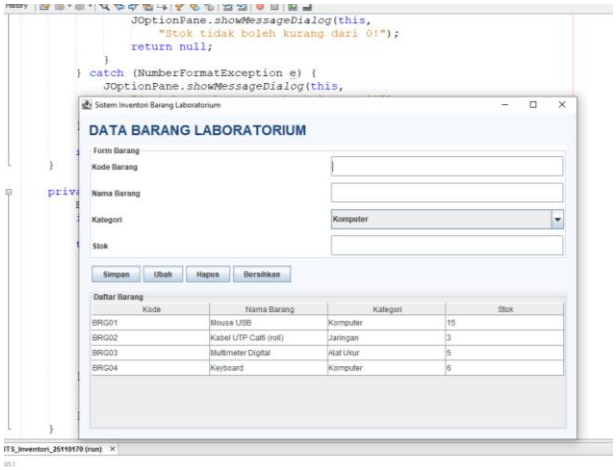
Memeriksa kode dan nama agar tidak kosong dan stok yang berupa angka >= 0. Jika data benar maka akan disimpan melalui class barang Dao dalam paket Dao dengan preparedStatement dan notifikasi berhasil atau gagal.

- Tampilan barang Awal

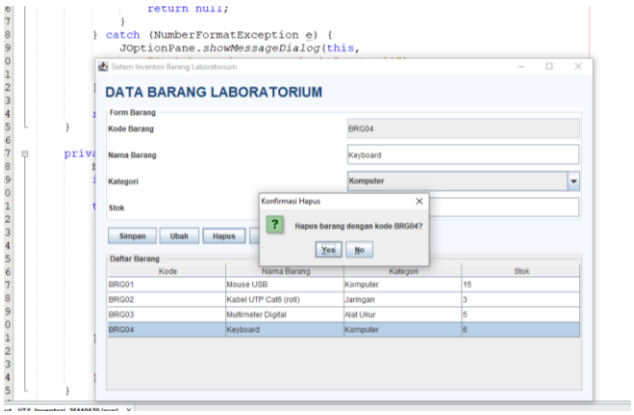


Dijalankan melalui class Form barang dari paket view

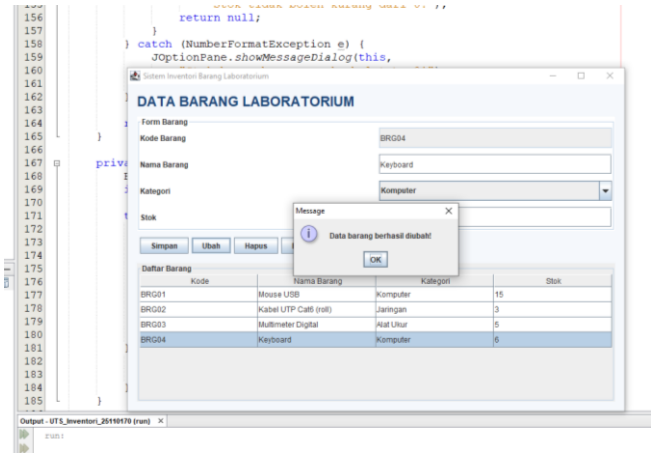
- Tampilan setelah data ditambahkan



- Tampilan saat data akan dihapus



- Tampilan Saat Ubah Data



- Uji Coba dengan mengisi stok dengan huruf

